

Brief Intro. to Deep Learning & Intro. of Textbook

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Special Lecture for Machine Learning

Outline

View of MLP

Connection with Brain

Universal Approximation and High variance

Deep Network

AI History

Deep Networks

Increasing Data and model Sizes

View of MLP I

1. Perceptron (Frank Rosenblatt)

- ▶ Binary classifier with a linear hyperplane, $\mathbb{I}(\mathbf{w}^T \mathbf{x} + b > 0)$.
- ▶ One layer with linear activation is the same as the linear classification (for simplicity, we omit the bias b)

초평면(hyperplane) 위/아래

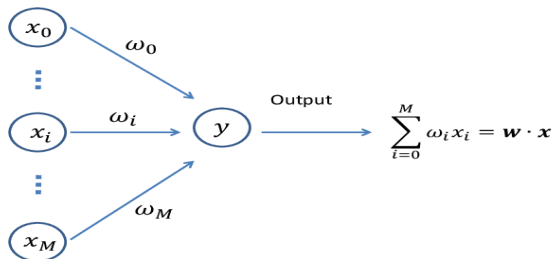


Figure 1: Heterogeneous sharpness for cross-spectral face recognition (2017), Zhicheng Cao, Natalia Schmid.

View of MLP II

► XOR problem

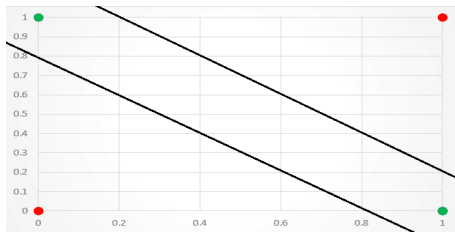


Figure 2: <https://dev.to/jbahire/demystifying-the-xor-problem-1blk>.

x	y	XOR
0	0	0
0	1	1
1	0	1
1	1	0

View of MLP III

- ▶ How to make a classifier to separate the classes? A non-linear decision boundary is required.

2. Multi-layer & Non-linear Activation

linear끼리 계산해도 linear

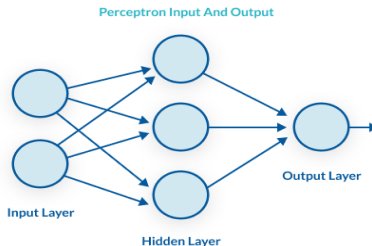


Figure 3:

<https://indiantechwarrior.com/perceptrons-and-multi-layer-perceptrons/>.

View of MLP IV

- ▶ Hidden layer: summation of linear functions and activation such as a sigmoid function such as $\sigma(x) = 1/(1 + \exp(-x))$.

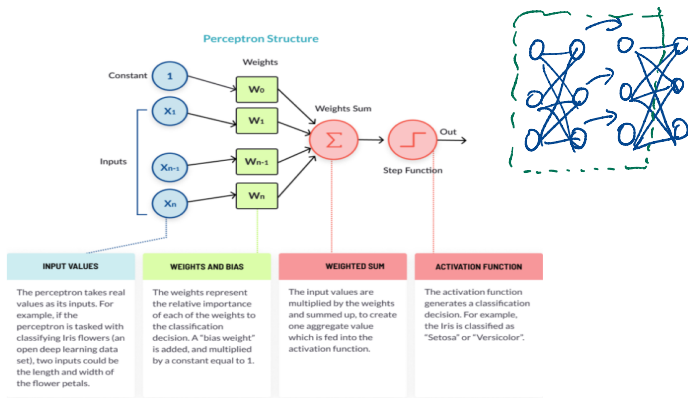


Figure 4: Details of perceptron structure

(<https://indiantechwarrior.com/perceptrons-and-multi-layer-perceptrons/>).

Connection with Brain I

1. Brain and Neuron

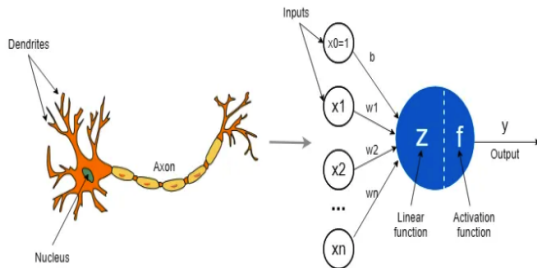


Image (at left) by [Ctker-Free-Vector-Images](#) from [Pixabay](#) and image (at right) by author, made with draw.io

Figure 5: <https://towardsdatascience.com/the-concept-of-artificial-neurons-perceptrons-in-neural-networks-fab22249cbfc>.

Connection with Brain II

- ▶ Perceptron is interpreted as the neuron in the brain having the connected biological organs playing input, activation, and output roles.
- ▶ Humans have 86 billion neurons (175 billion weights and biases in GPT3).
- ▶ Many studies have brain and deep-network-related topics in AI fields.

2. Neural Link

- ▶ Neural chip is used as two folders.
- ▶ Chip based on the biological principle.
- ▶ Planted chip on the brain.

Universal Approximation and High variance I

1. Universal Approximation Theorem

- ▶ George Cybenko(1989) proved that the continuous ftn. can be approximated by two-fully-connected-layers neural net.
- ▶ In the estimation, simple architecture is sufficient.
- ▶ This can bother the advance to DNN.

Let $C(X, \mathbb{R}^m)$ denotes the set of continuous function from a subset X of a Euclidean $\mathbb{R}^n$ space to a Euclidean space $\mathbb{R}^m$. Let $\sigma \in C(\mathbb{R}, \mathbb{R})$ where $(\sigma \circ x)_i = \sigma(x_i)$. Then σ is not polynomial if and only if for every $n, m \in \mathbb{N}$, compact $K \subset \mathbb{R}^n$, $f \in C(K, \mathbb{R}^m)$, $\epsilon > 0$, there exist $k \in \mathbb{N}$, $A \in \mathbb{R}^{k \times n}$, $b \in \mathbb{R}^k$, $C \in \mathbb{R}^{m \times k}$ such that

$$\sup_{x \in K} \|f(x) - g(x)\| < \epsilon$$

where $g(x) = C \cdot (\sigma \circ (Ax + b))$.

Universal Approximation and High variance II

DNN에서 High variance 해결

but unique 한 해는 X (randomness)

그러나 변동은 매우 작음

다층 layer를 쌓으면 생기는 문제점

2. High Variance

- ▶ NN is usually considered as having high variance due to the somewhat severe dependency on the initial values for parameters.
- ▶ This no-uniqueness is not solved in a recent DNN.
- ▶ Using the simple architecture is preferred temporarily.
- ▶ Instead of reducing the parameters, increasing parameters have been studied due to the potential of transformation.

Universal Approximation and High variance III

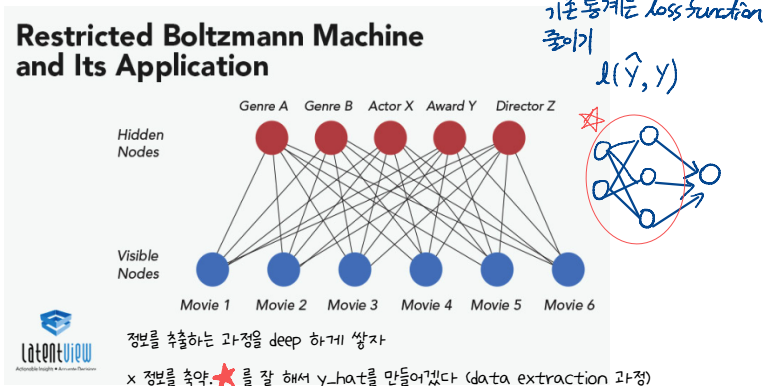


Figure 6: <https://www.latentview.com/blog/restricted-boltzmann-machine-and-its-application/>.

- ▶ RBM is to make a better representation (feature) of input for better-supervised learning.

Deep Network I

1. Innovations of Neural Networks

- ▶ History has many phases, such as Recurrent Neural Net (Rummelhart, 1986) and Boltzmann Machine (Hinton and Sejnowski, 1985).
- ▶ Back Prop. and multi-layer perceptron (Werbos, 1974; Rummelhart, Hinton & William, 1986)

Deep Network II

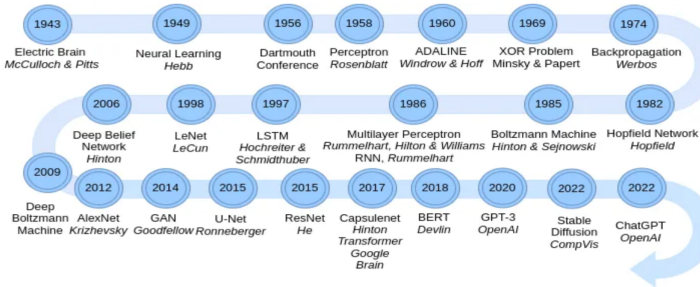


Image by Author

Figure 7: <https://pub.towardsai.net/a-brief-history-of-neural-nets-472107bc2c9c>.

Deep Network III

2. Deep

- ▶ Deep belief network (2006), Deep Boltzmann Machine (2009): unsupervised or representation learning with deep layers. Finding the potential and primitive clue for learning with deep layers.
- ▶ How to use the deep network for all-round players in machine learning? AlexNet (NeurIPS, 2017) provides the significant advance.

Deep Network IV

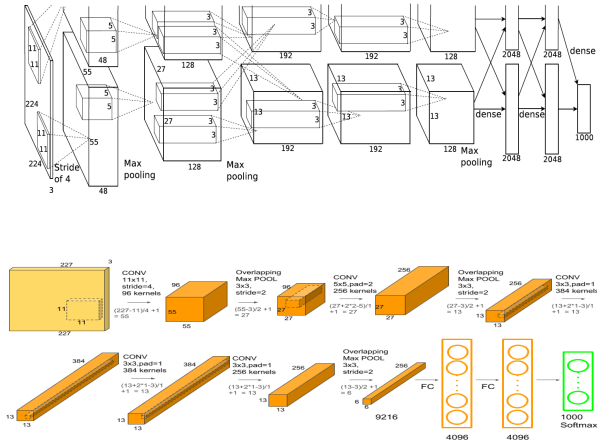


Figure 8: AlexNet (<https://www.kaggle.com/code/blurredmachine/alexnet-architecture-a-complete-guide>)

Deep Network V

► Innovations

1. Using ReLU (Rectified Linear Unit) to solve the gradient vanishing problem in the deep network.
2. CNN to reduce the computational cost in backpropagation.
3. Deep layers to achieve great performance.

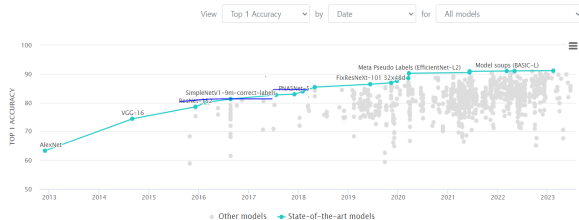


Figure 9: <https://paperswithcode.com/sota/image-classification-on-imagenet>

Outline

View of MLP

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Universal Approximation and High variance

Deep Network

AI History

Deep Networks

Increasing Data and model Sizes

AI history I

1. Before Machine Learning

- ▶ Creating machines that think this desire goes back to ancient Greece.
- ▶ Programmable computers were first conceived, and people wondered whether such machines might become intelligent.
- ▶ Many of the early successes of AI took place in relatively sterile and formal environments, not requiring computers to have much knowledge about the world.
- ▶ IBM's Deep Blue chess-playing system (2002) where the task is based on the formal rules.

AI history II

- ▶ Knowledge-based system
 1. Most bits of knowledge of human beings is not formal but intuitive.
 2. Coding the knowledge into computers (example: Cync).
 3. Non-successful system.

AI history III

2. Machine learning and representation

- ▶ Learning the real world is required based on data.
- ▶ SVM, naive Bayes, and others.
- ▶ (Simple) machine Learning algorithms depend heavily on the representation of the data they are given.
- ▶ Example

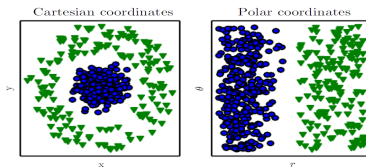


Figure 1.1: Example of different representations: suppose we want to separate two categories of data by drawing a line between them in a scatterplot. In the plot on the left, we represent some data using Cartesian coordinates, and the task is impossible. In the plot on the right, we represent the data with polar coordinates and the task becomes simple to solve with a vertical line. (Figure produced in collaboration with David Warde-Farley.)

Figure 10: Importance of features.

AI history IV

- ▶ Learned representations often result in much better performance than can be obtained with hand-designed representations.
- ▶ They also enable AI systems to rapidly adapt to new tasks with minimal human intervention.
- ▶ The quintessential example of a representation learning algorithm is the auto-encoder.
 1. Separating **factors of variation** that explained the observed data (unobserved or human mind).
 2. Most applications require us to disentangle factors of variation and discard the ones we do not care about.

AI history V

► Deep learning

1. Feedforward deep network, multilayer perceptron.
2. A multilayer perceptron is just a mathematical function mapping some input values to output values.
3. One view: learning the right representation for data
4. Second view: sequential instructions provide great power because later instructions can refer back to the results of earlier instructions.

AI history VI

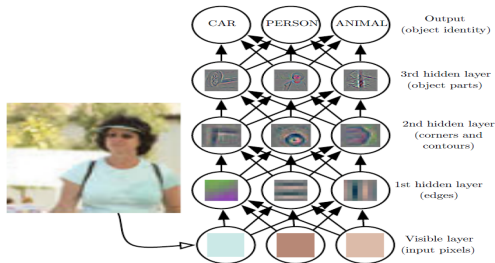


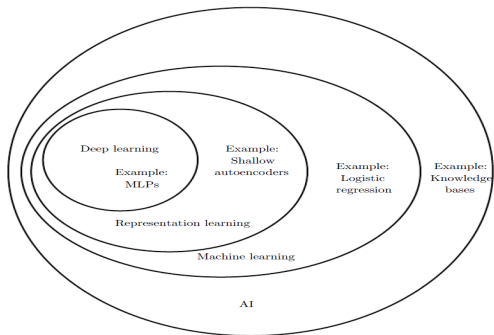
Figure 11: Representation and sequential instructions.

AI history VII

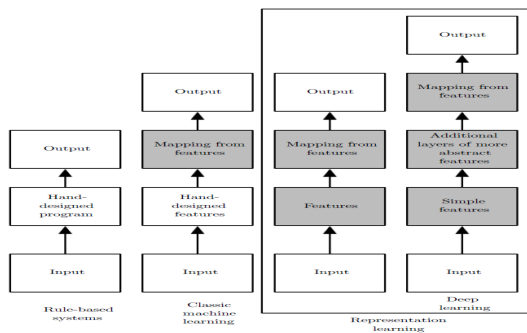
► Depth

1. Number of sequential instructions that must be executed to evaluate the architecture (can be different by the program type). Usually, the linear transformation with activation is referred to as “one layer”.
2. Probabilistic approach. For example, a layer for eyes and a layer for faces.

Deep Networks I



Deep Networks II



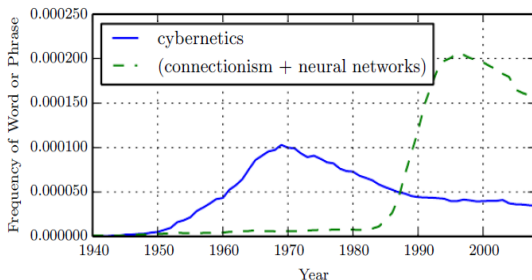
Deep Networks III

Historical Trends in Deep Networks

- ▶ Deep learning has had a long and rich history but has gone by many names, reflecting different philosophical viewpoints, and has waxed and waned in popularity.
- ▶ Deep learning has become more useful as the amount of available training data has increased.
- ▶ Deep learning models have grown in size over time as computer infrastructure (both hardware and software) for deep learning has improved.
- ▶ Deep learning has solved increasingly complicated applications with increasing accuracy over time.

Deep Networks IV

The Many Names and Changing Fortunes of Neural Networks



- Cybernetics (1940-60): biological modeling (learning the brain behaviors). The terminology of artificial neural networks (ANNs) arises.

Deep Networks V

- ▶ Neural net

1. A conceptually straightforward path to building intelligence is to reverse engineer the computational principles behind the brain and duplicate its functionality.
2. Understanding the brain and the principles that underlie human intelligence to be applied in machine learning.

- ▶ Neural net and neuroscience

1. Parallel in basic.
2. Deep networks are beyond the neuroscience, *more general principle of learning multiple levels of composition.*

Deep Networks VI

- ▶ Deep network and neuroscience
 1. Adaptive linear element (ADALINE) and SGD / multi-layer perceptron.
 2. There is evidence that much of the mammalian brain might use a single algorithm to solve most of the tasks that the brain solves. This implies that deep networks can be used in various machine-learning tasks in a unified manner.
 3. Neuroscience is an important source of inspiration.

Deep Networks VII

- ▶ Connection, parallel distributed processing (1)
 1. In 1980, the central idea in connectionism was that many simple computational units could achieve intelligent behavior when networked.
 2. Distributed representation. For example, color and shape factors for an image.
 3. In 1990, Researchers made important advances in modeling sequences with neural networks (RNN, LSTM)

Deep Networks VIII

- ▶ Connection, parallel distributed processing (2)
 1. Until 2006, impressive advances in computer vision appeared.
 2. At this point, deep networks were generally believed to be very difficult to train.

Deep Networks IX

- ▶ Deep networks

1. Deep belief, CNN with deep layers. This wave of neural network research popularized the term “deep learning” to emphasize that researchers could now train deeper neural networks than possible and focus attention on the theoretical importance of depth.
2. Outperforming all other AI systems based on other machine learning.

Increasing Data and model Sizes I

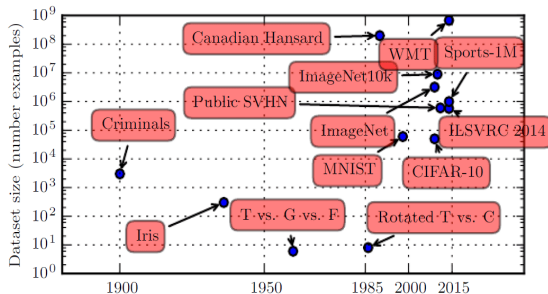
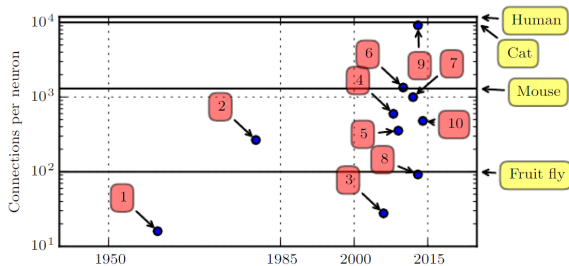


Figure 12: Benchmark dataset sizes along with years.

Increasing Data and model Sizes II



Increasing Data and model Sizes III

1. Adaptive linear element (Widrow and Hoff, 1960)
2. Neocognitron (Fukushima, 1980)
3. GPU-accelerated convolutional network (Chellapilla *et al.*, 2006)
4. Deep Boltzmann machine (Salakhutdinov and Hinton, 2009a)
5. Unsupervised convolutional network (Jarrett *et al.*, 2009)
6. GPU-accelerated multilayer perceptron (Ciresan *et al.*, 2010)
7. Distributed autoencoder (Le *et al.*, 2012)
8. Multi-GPU convolutional network (Krizhevsky *et al.*, 2012)
9. COTS HPC unsupervised convolutional network (Coates *et al.*, 2013)
10. GoogLeNet (Szegedy *et al.*, 2014a)

Figure 13: Model sizes along with years.

Increasing Data and model Sizes IV

- ▶ GPT3 data size: b690G
- ▶ GPT3 number of parameters: 175 billion